Sahaj Bhadja

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Summary

Software engineer with 2+ years building Python services, scalable data pipelines, REST APIs, and AI-enabled workflows. Experienced in ingestion, ETL, structured validation, SQL/NoSQL, AWS, orchestration, retries, observability, automated testing, and production reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python APIs and Azure SQL services for 500+ users, normalizing application data and tuning indexed queries and backend access paths to reduce production latency from 1.2 seconds to 350 milliseconds.
- Orchestrated 5,000+ monthly jobs through explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving pipeline observability, resumability, maintainability, artifact tracking, and production troubleshooting responsiveness.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Flask REST services handling 10,000+ daily requests, integrating secure document ingestion, authentication, structured status updates, and automated processing workflows while resolving production defects for 150+ users.
- Created multithreaded ingestion for 1,000+ files with schema validation, AWS RDS persistence, progress tracking, and failure recovery, reducing turnaround 60% while maintaining reliable behavior under concurrent production load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built testable PyTorch pipelines processing 100,000+ trajectory logs, improving planning accuracy 18% through reproducible experiments, structured model evaluation, performance monitoring, validation checks, and systematic debugging of AI-generated behavior.
- Developed a spatial-temporal encoder combining map geometry and agent context through multi-head attention, improving closed-loop stability 15% while documenting evidence and validating outputs across diverse simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed Python services, source integrations, and ETL workflows using AWS Glue, S3, SQL, and REST APIs, improving response times 30% and data-processing efficiency 68% across production cloud applications.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito, accelerating schema validation, defect diagnosis, release verification, and reliable deployments across Windows, Linux, and macOS.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular Node.js services across eight microservices with authentication, role-based access, WebSocket messaging, and state synchronization, coordinating 2 million+ Azure SQL records through clearly bounded secure interfaces.
- Automated Docker builds, functional tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving service quality, deployment consistency, and operational visibility.

Skills

Languages & Services: Python, Java, JavaScript, TypeScript, SQL, Node.js, Flask, FastAPI, REST APIs, microservices, WebSockets, service boundaries
Data & AI: ETL, data ingestion, schema validation, data modeling, PostgreSQL, Azure SQL, MySQL, MongoDB, LLaMA-2, RAG, FAISS, semantic search, prompt engineering
Cloud & Pipelines: AWS, Azure, AWS Glue, S3, AWS RDS, Blob Storage, Docker, Kubernetes, job orchestration, state tracking, retries, failure recovery, CI/CD
Quality & Delivery: Git, Jenkins, PyTest, Mockito, unit testing, integration testing, automated testing, debugging, logging, monitoring, observability, code review, documentation, production support, Agile

Projects

Big-Brain | Live | *LLaMA-2, RAG, FAISS, PostgreSQL, REST APIs, Clerk*
- Built an authenticated RAG application processing 50,000+ unstructured documents with LLaMA-2, FAISS, PostgreSQL, and REST APIs, combining semantic retrieval, conversational context, access controls, and real-time collaboration workflows.

Chat2SQL | Code | *Java, TypeScript, SQL, PostgreSQL, MySQL, MongoDB, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases with schema-aware query generation, structured validation, optimized execution, metadata exploration, and Docker-backed deployment across modular Java and TypeScript services.

FinNexus | Live | *TypeScript, REST APIs, WebSockets, MongoDB, Kubernetes, CI/CD*
- Built a scalable platform supporting 50,000+ users and one million transactions, integrating secure APIs, real-time synchronization, MongoDB persistence, Kubernetes deployment, operational monitoring, and automated CI/CD release workflows.